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# Alterations in performance and discriminating power of the death/suicide implicit association test across the lifespan

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## Highlights

- Suicide is a major public health concern in the United states.
- Death/suicide implicit associations shown to be objective marker for suicide risk.
- Age effect associated with key measures on death/suicide implicit association test.
- Death/suicide implicit associations differentiated suicide risk only in younger ages.

## Abstract

The Death/Suicide Implicit Association Test (d/s-IAT) has differentiated individuals with prior and prospective suicide attempts in previous studies, however, age effects on test results remains to be explored. A three-site study compared performance on the d/s-IAT among participants aged 16–80 years with depression and prior suicide attempt ($n$ = 82), with depression and no attempts ($n$ = 80), and healthy controls ($n$ = 86). Outcome measures included the standard difference ($D$) score, median reaction times, and error rates. Higher $D$ scores represent a stronger association between

death/suicide and self, while lower scores represent a stronger association between life and self. The *D* scores differed significantly among groups overall. Participants with depression exhibited higher scores compared to healthy controls, but there was no difference between participants with and without prior suicide attempts($F[2,242]=8.76$, $p<.001$). Response times for participants with prior attempts differed significantly from other groups, with no significant differences in error rates. The *D* score was significantly affected by age ($\beta$ =-0.007, $t$ = 3.65, $p<.001$), with slowing of response times in older ages. Results suggest reaction time d/s-IAT *D* scores may not distinguish implicit thinking about suicide as response times slow with age, but slowed response times may be sensitive to suicide risk potentially indicating basic information processing deficits.

## Introduction

Suicide is a major public health concern in the United States, accounting for over 45,000 deaths in 2020 (CDC, 2020). To support their vision for a “nation free from the tragic experience of suicide,” the National Action Alliance for Suicide Prevention (NAASP) established a research agenda to reduce the annual suicide rate 20 % by 2025 (NAASP, 2014). This agenda includes identifying improved capabilities to better detect individuals at highest risk for suicide across care settings. An existing limitation in our ability to effectively assess suicide risk is the almost exclusive reliance on self-report (Franklin et al., 2017; Kleiman et al., 2017). This can be especially problematic due to the transient nature and potential reluctance of individuals to express suicidal thoughts and behaviors (Carter et al., 2017; Franklin et al., 2017; Nock and Banaji, 2007). Detecting implicit associations with death and suicide has been offered as a more objective marker of suicide risk that can provide a complementary approach to patient self-report (Glenn et al., 2017a, b; Millner et al., 2018; Ellis et al., 2016).

The most widely studied measure of these implicit associations is the Death/Suicide Implicit Association Test (d/s-IAT) (Barnes et al., 2017; Nock et al., 2010; Sohn et al., 2021). The d/s-IAT examines implicit self-identification with death versus life by evaluating the relative difference in response times when life vs. death/suicide related stimuli are associated with the self (Nock et al., 2010; Nock and Banaji, 2007). Prior research has demonstrated that individuals at risk for suicide respond relatively more quickly than non-suicidal individuals to stimuli associating “death/suicide” and “me”, than when “life” and “me” are paired (Barnes et al., 2017; Glenn et al., 2017a,b; Nock et al., 2010; Tello et al., 2020). In a study by Nock et al. (2010), adults presenting to a psychiatric emergency department following a suicide attempt had relatively faster implicit associations pairing the self with death/suicide compared to those with other psychiatric emergencies. These patients also had a six-fold increase in the risk for a suicide attempt during the 6-month follow-up period (Nock et al., 2010). Similar findings were obtained in veterans in a psychiatric inpatient setting (Barnes et al., 2017), with veterans expressing relatively faster associations between self and death (compared to self and life) twice as likely to attempt suicide within 6 months of the baseline assessment (Barnes et al., 2017).

Evidence regarding the d/s-IAT and suicide risk has been inconsistent. In a study by Rath et al. (2021), d/s-IAT scores were not predictive of suicide attempts in adult psychiatric inpatients above and beyond depression, prior suicide attempts, and suicidal ideation. Glenn et al. (2019) found the d/-s-IAT was significantly associated with future suicide ideation in youth aged 12–19 years

receiving psychiatric outpatient treatment, but not when controlling for baseline ideation. An additional study by Brent et al. (2021) among youth in pediatric emergency departments found the d/s-IAT to be only a modest predictor of future suicide attempts, with self-report of known suicidal risk factors showing greater predictive accuracy.

While research involving the d/s-IAT has been conducted with mostly younger populations, or populations within a relatively restricted age range (Barnes et al., 2017; Brent et al., 2021; Nock et al., 2010; Glenn et al., 2019), little is known about the discriminating power of the d/s-IAT in older adults. The current study examined whether implicit associations with death/suicide distinguished participants with depression and prior suicide attempts, participants with depression and no attempts, and healthy controls across the adult lifespan. Based on existing literature, we hypothesized that participants with prior suicide attempts would have higher difference scores on the d/s-IAT (faster relative association between self and death than to self and life) compared to participants with no attempts and healthy controls, and that participants with no attempts would have higher difference scores relative to healthy controls. We also sought to determine how age itself might affect the association between group status and d/s-IAT performance. The IAT outcome measure relies on reaction time differences between different conditions, and it is possible that age effects on reaction times may have an impact on the discriminating power of these scores. The present study examined participants aged 16 to 80 years, allowing for analysis of these aging effects, in addition to standard group differences.

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## Section snippets

### Sample

Participants were recruited from three large metropolitan university hospitals and affiliated health care providers in Pittsburgh, PA, Columbus, OH, and New York City, NY as part of a collaborative multi-center study to examine age associations with clinical and neurocognitive risk factors for suicidal behavior. Exclusion criteria included a history of neurological disorder, delirium, psychosis, mania, dementia, or current substance abuse disorder. Individuals who scored <24 on the Mini-Mental …

### Demographics and clinical ratings

The study sample was predominantly female, White, and non-Hispanic, with an overall average age of 41.8 ± 17.4 (range from 16 to 80) with no differences among the clinical groups. Years of education ($F[2245]=6.31$, $p=.002$), and MMSE scores ($F[2245]=3.63$, $p=.028$) were significantly lower among participants with a history of suicide attempts than healthy controls, but not compared to

those with no attempts (Table 1). Estimated intellectual ability was comparable across groups, and within one point …

## Discussion

Previous studies have demonstrated the predictive value of the d/s-IAT for suicide risk (Barnes et al., 2016; Ellis et al., 2016; Nock et al., 2010), but age effects appear to impact its discriminating power. In this cross-sectional adult lifespan sample, age was found to have a substantial negative association with the magnitude of the d/s-IAT *D* score. Our findings indicate that *D* scores no longer identified those with a history of suicide attempt in middle-aged and older participants. The …

## Limitations

The cross-sectional design of this study limits conclusions regarding individual developmental trajectories on the d/s-IAT. Second, most of the sample was female, White and non-Hispanic. Future studies should include greater representation of males and other racial and ethnic groups. Third, a higher percentage of individuals with more recent suicidal behavior, within the last six months to a year, may have resulted in stronger group differences, consistent with existing research (Glenn et al., …

## Conclusion

The d/s-IAT administered in this study did not show group differences in past suicidal behavior at middle and older ages, primarily because the *D* score was sensitive to the slowing of reaction times with increasing age. Effects consistent with prior research were found in younger participants in this study and those with more recent suicidal behavior, but not across the full age span sampled here. Alternative methods of computing the *D* score may be needed to reduce its susceptibility to overall …

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## CRediT authorship contribution statement

**Donna A. Ruch:** Conceptualization, Methodology, Formal analysis, Validation, Writing – original draft. **Jeffrey A. Bridge:** Funding acquisition, Supervision, Conceptualization, Methodology, Writing – review & editing. **Jaclyn Tissue:** Project administration, Data curation, Writing – review & editing. **Sean P. Madden:** Writing – review & editing. **Hanga Galfavy:** Methodology, Formal analysis, Writing – review & editing. **Marianne Gorlyn:** Writing – review & editing. **Arielle H. Sheftall:** Data curation, …

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. …

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